

ETHAN C. DONLON

Inspired Multidisciplinary Robotics Engineer specializing in full-stack mechatronic systems. Equally fluent in software, electronics, and mechanical design. Proven ability to own complex systems from concept through production, bringing an unparalleled work ethic to solving challenges at the intersection of hardware and software.

EXPERIENCE

Pickle Robot Company, Cambridge, MA

Mechatronics Engineer IV

Mechatronics Engineer

Mechatronics Engineer – Junior Member of Technical Staff+

Mechatronics Engineer – Junior Member of Technical Staff

November 2022 – Current

(September 2025 – Current)

(March 2025 – September 2025)

(February 2024 – March 2025)

(November 2022 – February 2024)

- Joined the company at Series A and contributed to scaling engineering, customer deployments, and technical execution that supported a successful \$50M Series B raise and nationwide customer adoption across the United States
- Implemented custom communication drivers using the Berkeley Sockets API (Python) over UDP and TCP for robot and PLC control; built internal APIs supporting higher-level interfaces within a large ROS-based codebase
- Developed embedded software for real-time control of KUKA robot arms and a simulated teach pendant
- Designed and implemented a custom bare-metal STM32 microcontroller-based gripper, complete with an ethernet switch and IO-Link firmware in C, eliminating \$20,000 in licensing costs
- Designed and deployed a 2D LiDAR-based human presence detection system to achieve functional safety compliance
- Owned system-level sensor and controls integration, including architecture definition, signal routing, and low-level communication interfaces across CAN-based motor controllers, PROFISAFE, IP-based devices, and discrete I/O
- Led cross-functional debugging efforts for motor controllers, sensors, communication networks, and safety systems
- Owned robot PLC programs and developed a Linux command-line tool for remote PLC deployment by reverse engineering a third-party protocol, eliminating onsite Windows-based updates
- Designed, prototyped, and delivered the Gen2 test-bed platform featuring multiple control panels and custom PCBs under aggressive timelines
- Led development of the Gen1 electrical system, reducing per-robot build cost from \$5,600 to \$1,000 while improving reliability and manufacturability through custom PCB designs
- Architected and implemented a single 480V power input system for the Gen1 system, saving customers approximately \$20,000 in electrical installation costs per deployment
- Designed Pickle's first electrical cabinet for the Gen0 system, achieving zero field electrical failures across all deployed Gen0 systems - learned by reading 100's of pages of UL, IEC, and NFPA Standards
- Established CM partnerships and authored detailed production and test documentation to scale deployment

DOVE Opioid Device, AltruMed, Startup at the University of Pennsylvania

Founding Engineer

January 2021 – September 2022

(January 2021 – September 2022)

- Designed and implemented electromechanical systems and embedded firmware on custom PCBs for a wearable medical device capable of sensing hypoxia and automatically administering naloxone
- Supported early clinical study planning and in field research
- Awarded the Coulter Grant for innovation addressing underserved healthcare needs

Deka Research and Development, Manchester, NH

Mechanical Engineering Intern

Manufacturing Intern

June 2020 – August 2021

(June 2021 – August 2021)

(June 2020 – August 2020)

- Completed design changes, complex three-dimensional tolerance analyses, and manufacturing fixture development for a Class III medical device

EDUCATION

University of Pennsylvania, Philadelphia, PA

Master's in Mechanical Engineering, Concentration in Mechatronics, **GPA: 3.8**

September 2019 – May 2022

Haverford College, Haverford, PA

Bachelor of Science in Physics with High Honors, Minor in German, *magna cum laude*, **GPA: 3.94**

September 2017 – May 2021

Study Abroad, Berlin, Germany

Humboldt University; German Business (B2) Proficient

June 2019 – August 2019

TECHNICAL SKILLS

(proficient) Python, Git, Unix, PLC Programming, Kuka Robot Language, Wireshark, Altium, KiCad, Onshape, Solidworks, Oscilloscopes, Power Tools (familiar) ROS, Docker, C, C++, JavaScript, HTML/CSS, can learn anything